

Document Control

- System Name: Application
- Version: 1.0
- Date: Not present in source JSON
- Author: Not present in source JSON

Executive Summary

- This document details business requirements for ingesting customer and order CSV data into Delta tables.
- System Name: Application
- Objectives cover data cleansing, transformation, SCD Type 2 handling, aggregation, and implementation using Delta Live Tables.
- Strict adherence to names, schemas, and rules from the source requirements.

Business Objectives

- Load customer and order CSVs into Delta tables.
- Cleanse data by removing "Null"/null literals and duplicates.
- Enrich order data by computing $\text{TotalAmount} = \text{PricePerUnit} \times \text{Qty}$.
- Join customer and order to create an ordersummary table.
- Apply SCD Type 2 handling for customer changes in ordersummary.
- Aggregate daily customer spend in customeraggregatespend table.
- Complete the full pipeline using Delta Live Tables.

In-Scope

- Ingestion from CSV sources at specified filesystem paths.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Execution of data cleansing and computation activities as described.
- SCD Type 2 processing for customer history in ordersummary.
- Aggregation process into customeraggregatespend.
- Implementation via Delta Live Tables.

Out-of-Scope

- Any sources, schemas, fields not explicitly described.
- Security, access controls, SLAs, monitoring, alerting.
- Performance, scaling, and scheduling beyond specified Delta Live Tables operations.
- Data quality metrics beyond listed cleansing actions.

- User interfaces, reports, or analytics layers.

Stakeholders

- Not present in source JSON

Business Requirements

BR-APP-001 — Source Ingestion

- Description: Load customer and order source CSV data into Delta tables from designated filesystem locations.
- Business Value: Establishes foundational datasets for downstream processing and analytics.
- Priority: High

BR-APP-002 — Table Schema Compliance

- Description: Create and enforce schemas for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) tables.
- Business Value: Ensures data consistency and accurate processing.
- Priority: High

BR-APP-003 — Computed Field TotalAmount

- Description: Calculate TotalAmount on the order table as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Supports downstream financial and aggregation analytics.
- Priority: High

BR-APP-004 — Data Cleansing Operations

- Description: Remove "Null"/null literals and duplicate records from customer and order tables.
- Business Value: Maintains data quality and reliability.
- Priority: High

BR-APP-005 — Ordersummary Target Table Creation

- Description: Create ordersummary table in specified catalog/schema, enabling historical and joined customer-order data.
- Business Value: Provides centralized view for reporting and SCD processing.
- Priority: High

BR-APP-006 — SCD Type 2 Change Handling for customer

- Description: Set previous customer records inactive and new records active upon changes; update StartDate and EndDate; maintain full history in ordersummary.
- Business Value: Ensures historical customer changes are auditable.

- Priority: High

BR-APP-007 — Join Logic

- Description: Join customer and order tables on CustId to populate ordersummary.
- Business Value: Integrates customer identity with transaction data.
- Priority: High

BR-APP-008 — Aggregation Processing

- Description: Aggregate data from ordersummary to customeraggregatespend table by Name and Date, summing TotalAmount.
- Business Value: Delivers summarized spend data for analysis.
- Priority: High

BR-APP-009 — Implementation via Delta Live Tables

- Description: Execute all pipeline steps using Delta Live Tables as per requirements.
- Business Value: Leverages managed data pipelines for reliability and automation.
- Priority: High

Assumptions & Constraints

- All implementation utilizes Delta Live Tables.
- No business logic beyond what is explicitly specified.
- Any fields required for SCD processing (StartDate, EndDate, Active) are referenced but not explicitly defined in the schema.
- No assumptions beyond those explicitly stated or inferable from requirements.

Success Metrics

- Delta tables customer, order, ordersummary, and customeraggregatespend are accurately created.
- Data cleansing removes all "Null"/null and duplicate values.
- Computed TotalAmount is present and correct in order and customeraggregatespend tables.
- SCD Type 2 history is correctly maintained in ordersummary.
- Daily aggregates in customeraggregatespend accurately reflect underlying transactions.

Risks & Mitigations

- Risk: Lack of explicit field definitions for SCD Type 2 attributes.
- Mitigation: Use industry-standard field names where possible, log open questions.
- Risk: Data quality issues beyond cleansing.
- Mitigation: Strict cleansing; additional issues deferred to future work.

- Risk: Monitoring and error handling not addressed in scope.
- Mitigation: Mark as potential open requirement for future iterations.

Approval

- Approver Name: Not present in source JSON
- Date: Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Bronze	customer	CustId, Name, EmailId, Region	Customer master
	Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Order Delta table
	Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Join of customer
nd	Gold	customeraggregatespend	Name, TotalAmount, Date	Aggregate spend

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Load CSV	Customer/Order source paths	customer/order tables	As CSV-to-Delta load
2	Cleansing	customer/order	cleansed tables	Remove "Null"/null literals, deduplicate
3	Computed Column	order	order with TotalAmount	TotalAmount = PricePerUnit × Qty
4	Join & SCD	customer + order	ordersummary	Join on CustId, apply SCD Type 2
5	Aggregation	ordersummary	customeraggregatespend	Group by Name, Date; sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR1	Computation	TotalAmount = PricePerUnit × Qty	Requirements
BR2	Cleansing	Remove "Null"/null literals, deduplicate	Requirements
BR3	Join	Join on CustId	Requirements
BR4	SCD Type 2	Change handling: Start/EndDate, Active	Requirements
BR5	Aggregation	Sum by Name, Date	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer/order CSV	None
2	Cleanse data	Step 1
3	Compute TotalAmount	Step 2
4	Join for ordersummary, SCD logic	Step 3
5	Aggregate to customeraggregatespend	Step 4

Appendix F: Data Pipeline Architecture Diagram

